VIT

Vellore Institute of Technology

Reg. Number 22BAI1266

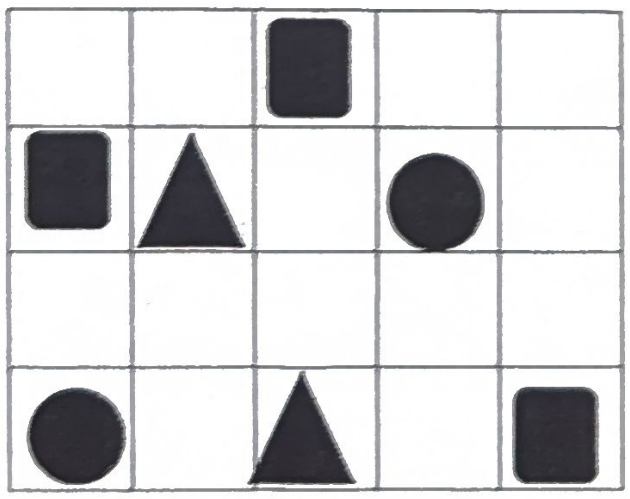
Continuous Assessment Test (CAT) –II - MARCH 2025

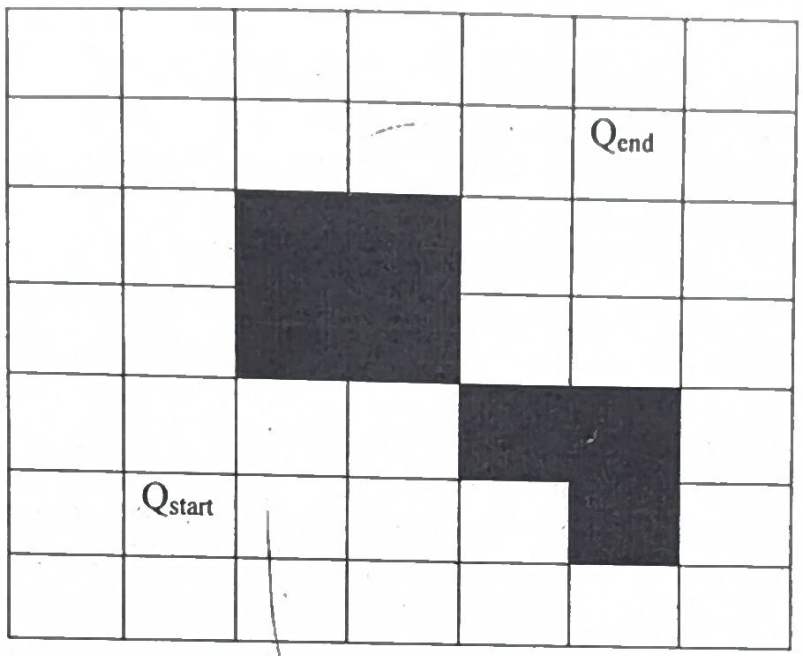
Programme	:	B.Tech. CSE (AIR)	Semester	:	WINTER SEMESTER 2024-25
Course Code & Course Title	:	BCSE427L Cognitive Robotics	Class Number	:	CH2024250502615 CH2024250502394
Faculty	:	Dr. S. Harini Dr.V.Arunkumar	Slot	:	D1
Duration	:	90 minutes	Max. Mark		50

General Instructions:

- Write only your registration number on the question paper in the box provided and do not write other information.
- Only non-programmable calculator without storage is permitted

Answer all questions

Q. No	Sub Sec.	Description	Marks
1.	a)	Design a neural network-based intelligence system for a mobile robot conducting search and rescue operations across different terrain types. The robot must continuously navigate various terrains, capture image feeds, and identify objects in its environment. Let 'n' represent the number of terrain types and 'm' represents the number of objects the system can identify Provide an overview of your neural network design, explaining how the system enables the robot to capture images, and recognize objects. Include pseudo-code to illustrate the approach. (10 marks)	16
	b)	Describe the step-by-step implementation of your design, using an example to demonstrate its functionality. (6 marks)	
2.		 Figure 1. Overview of the environment with 3 types of obstacles with 2 occurrences each	20

	a)	For the environment given in Figure 1, give the occupancy grid details assuming the robot is in third row, third column. The robot has started exploring the grids from the cell (0,0) and has explored only till (3,3).	
	b)	Explain your occupancy grid details(6 marks) Can you apply feature based map representation of the environment in Figure 1? Justify your answer in detail (6 marks)	
	c)	Assuming each cell is a square of side 1 cm, apply the procedures of the depth first map building algorithm for the Figure 1 environment and build a map. Explain the step by step working of the same with neat illustration. (8 marks)	
3.		 Figure 2. Overview of an environment with obstacles In figure 2, obstacles are the cells filled in black. The robot's starting and ending position are given as Q_{start} and Q_{end} respectively.	14
	a)	Apply sweep-line algorithm and build the visibility graph of the environment in Figure 2. Explain all the steps in detail in the application. (7 marks)	
	b)	For the environment in Figure 2, apply the Generalized Voronoi Diagram and calculate the cell distance by cellular automata. Illustrate your answer with a neat sketch of the grid cells and a detailed explanation. (7 marks)	
		Total	50

**********All the best **********